Introduction to the MIT Press Edition

This edition of *What Computers Can't Do* marks not only a change of publisher and a slight change of title; it also marks a change of status. The book now offers not a controversial position in an ongoing debate but a view of a bygone period of history. For now that the twentieth century is drawing to a close, it is becoming clear that one of the great dreams of the century is ending too. Almost half a century ago computer pioneer Alan Turing suggested that a high-speed digital computer, programmed with rules and facts, might exhibit intelligent behavior. Thus was born the field later called artificial intelligence (AI). After fifty years of effort, however, it is now clear to all but a few diehards that this attempt to produce general intelligence has failed. This failure does not mean that this sort of AI is impossible; no one has been able to come up with such a negative proof. Rather, it has turned out that, for the time being at least, the research program based on the assumption that human beings produce intelligence using facts and rules has reached a dead end, and there is no reason to think it could ever succeed. Indeed, what John Haugeland has called Good Old-Fashioned AI (GOFAI) is a paradigm case of what philosophers of science call a degenerating research program.

A degenerating research program, as defined by Imre Lakatos, is a scientific enterprise that starts out with great promise, offering a new approach that leads to impressive results in a limited domain. Almost inevitably researchers will want to try to apply the approach more broadly, starting with problems that are in some way similar to the original one. As long as it succeeds, the research program expands and attracts followers. If, however, researchers start encountering unexpected but important phenomena that consistently resist the new techniques, the program will stagnate, and researchers will abandon it as soon as a progressive alternative approach becomes available.

We can see this very pattern in the history of GOFAI. The program began auspiciously with Allen Newell and Herbert Simon's work at RAND. In the late 1950s Newell and Simon proved that computers could do more than calculate. They demonstrated that a computer's strings of bits could be made to stand for anything, including features of the real world, and that its programs could be used as rules for relating these features. The structure of an expression in the computer, then, could represent a state of affairs in the world whose features had the same structure, and the computer could serve as a physical symbol system storing and manipulating such representations. In this way, Newell and Simon claimed, computers could be used to simulate important aspects of intelligence. Thus the information-processing model of the mind was born.

Newell and Simon's early work was impressive, and by the late 1960s, thanks to a series of micro-world successes such as Terry Winograd's SHRDLU, a program that could respond to English-like commands by moving simulated, idealized blocks (see pp. 12–13), AI had become a flourishing research program. The field had its Ph.D. programs, professional societies, international meetings, and even its gurus. It looked like all one had to do was extend, combine, and render more realistic the micro-worlds and one would soon have genuine artificial intelligence. Marvin Minsky, head of the M.I.T. AI project, announced: "Within a generation the problem of creating 'artificial intelligence' will be substantially solved."[1]

Then, suddenly, the field ran into unexpected difficulties. The trouble started with the failure of attempts to program an understanding of children's stories (see pp. 57–62). The programs lacked the common sense of a four-year-old, and no one knew how to give them the background knowledge necessary for understanding even the simplest stories. An old rationalist dream was at the heart of the problem. GOFAI is based on the Cartesian idea that all understanding consists in forming and using appropriate symbolic representations. For Descartes, these representations were complex descriptions built up out of primitive ideas or elements. Kant added the important idea that all concepts are rules for relating such elements, and Frege showed that rules could be formalized so that they could be manipulated without intuition or interpretation. Given the nature of computers as possible formal symbol processors, AI turned this rationalist vision into a research program and took up the search for the primitives and formal rules that captured everyday knowledge. Commonsense understanding had to be represented as a huge data structure comprised of facts plus rules for relating and applying those facts. As it turned out, though, it was much harder than anyone expected to formulate, let alone formalize, the required theory of common sense. It was not, as Minsky had hoped, just a question of cataloging 10 million facts. Minsky's mood changed completely in the course of fifteen years. In 1982 he told a reporter: "The AI problem is one of the hardest science has ever undertaken."[2]

My work from 1965 on can be seen in retrospect as a repeatedly revised attempt to justify my intuition, based on my study of Martin Heidegger, Maurice Merleau-Ponty, and the later Wittgenstein, that the GOFAI research program would eventually fail. My first take on the inherent difficulties of the symbolic information-processing model of the mind was that our sense of relevance was holistic and required involvement in ongoing activity, whereas symbol representations were atomistic and totally detached from such activity. By the time of the second edition of *What Computers Can't Do* in 1979, the problem of representing what I had vaguely been referring to as the holistic context was beginning to be perceived by AI researchers as a serious obstacle. In my new introduction I therefore tried to show that what they called the commonsense-knowledge problem was not really a problem about how to represent *knowledge*; rather, the everyday commonsense background understanding that allows us to experience what is currently relevant as we deal with things and people is a kind of *know-how*. The problem precisely was that this know-how, along with all the interests, feelings, motivations, and bodily capacities that go to make a human being, would have had to be conveyed to the computer as knowledge—

as a huge and complex belief system—and making our inarticulate, preconceptual background understanding of what it is like to be a human being explicit in a symbolic representation seemed to me a hopeless task.

For this reason I doubted that the commonsense-knowledge problem could be solved by GOFAI techniques, but I could not justify my suspicion that the know-how that made up the background of common sense could not itself be represented by data structures made up of facts and rules. Granted that our background knowledge consists largely of skills for dealing with things and people rather than facts about them, what I needed was an argument against those who assumed that such skills were representable in symbolic form. As it turned out, my brother Stuart provided the missing argument in his phenomenological account of skill acquisition.[3]

Skill acquisition, he pointed out, usually begins with a student learning and applying rules for manipulating context-free elements. This is the grain of truth in the information-processing model. Thus a beginner at chess learns to follow strict rules relating such features as center control and material balance. After one begins to understand a domain, however, one sees meaningful aspects, not context-free features. Thus the more experienced chess player sees context-dependent characteristics such as unbalanced pawn structure or weakness on the king side. At the next stage, a competent performer learns to set goals and then look at the current situation in terms of what is relevant to achieving those goals. A further stage of proficiency is achieved when, after a great deal of experience, a player is able to see a situation as having a certain significance tending toward a certain outcome, and certain aspects of the situation stand out as salient in relation to that end. Given an appropriate board position, for example, almost all masters would observe after a few seconds of examination that to win white must attack the king side.

Finally, after even more experience, one reaches the level where one sees immediately what must be done. A chess grandmaster, for example, not only sees the issues in a position almost immediately, but the right response just pops into his or her head. There is no reason to suppose that the beginner's features and rules, or any other features and rules, play any role in such expert performance.[4] That we once followed a rule in learning to tie our shoelaces does not show, as Edward Feigenbaum argues it does,[5] that we must still be following that same rule unconsciously whenever we tie a lace. That would be like claiming that since we needed training wheels when learning how to ride a bicycle, we must now be using invisible training wheels whenever we ride. There is no reason to think that the rules that play a role in the *acquisition* of a skill play a role in its later *application*.

When *Mind Over Machine* came out, however, Stuart and I faced the same objection that had been raised against my appeal to holism in *What Computers Can't Do*. You may have described how expertise *feels*, critics said, but our only way of *explaining* the production of intelligent behavior is by using symbolic representations, and so that must be the underlying causal mechanism. Newell and Simon resort to this type of defense of symbolic AI:

> The principal body of evidence for the symbol-system hypothesis . . . is negative evidence: the absence of specific competing hypotheses as to how intelligent activity might be accomplished whether by man or by machine.[6]

In order to respond to this "what else could it be" defense of the physical symbol system research program, we appealed in *Mind Over Machine* to a somewhat vague and implausible idea that the brain might store holograms of situations paired with appropriate responses, allowing it to respond to situations in ways it had successfully responded to similar situations in the past. The crucial idea was that in hologram matching one had a model of similarity recognition that did not require analysis of the similarity of two patterns in terms of a set of common features. But the model was not convincing. No one had found anything resembling holograms in the brain.

At this point, like Charlie Chaplin in *Modern Times* emerging from a manhole with a red flag just as the revolutionaries came swarming by, we happily found ourselves surrounded by the rapidly growing ranks of neural-network modelers. As the commonsense-knowledge problem continued to resist the techniques that had worked so well in problem solving, and as pattern recognition and learning turned out to be much more intractable than anticipated, this alternative way of using computers to produce intelligence reemerged as an attractive research program after a long period of dormancy. The triumphant arrival of the neural-net revolutionaries, also called connectionists, completed the degeneration of the GOFAI research program.

The proposal that we should set about creating artificial intelligence by modeling the brain's learning power rather than the mind's symbolic representation of the world drew its inspiration not from philosophy but from what was soon to be called neuroscience. It was directly inspired by the work of D. O. Hebb, who had suggested in 1949 that a mass of neurons could learn if the simultaneous excitation of neuron A and neuron B increased the strength of the connection between them.[7] This lead was followed in the late 1950s by Frank Rosenblatt, who reasoned that since it was probably going to be hard to formalize intelligent behavior, AI should instead attempt to automate the procedures by which a network of neurons learns to discriminate patterns and respond appropriately. Researchers seeking symbolic representations were looking for a formal structure that would give computers the ability to solve a certain class of problems or discriminate certain types of patterns. Rosenblatt, conversely, wanted to build a physical device, or simulate such a device on a digital computer, that could generate its own abilities.

When symbolic AI seemed to stall, Donald Norman's Parallel Distributed Processing group and others started investigating variations of Rosenblatt's project and chalked up surprising successes. Soon, frustrated AI researchers, tired of clinging to a research program that Jerry Lettvin characterized in the early 1980s as "the only straw afloat," began defecting to the revived paradigm. Rumelhart, McClelland, and the PDP Research Group's two-volume work, *Parallel Distributed Processing*, had 6000 backorders the day it went on the market in 1986, and over 45,000 sets are now in print. Like the dissolution of the Soviet Union, the speed of collapse of the GOFAI research program has taken everyone, even those of us who expected it to happen sooner or later, by surprise.[8]

Happily for Stuart and me, the neural-network modelers had a much more plausible answer to the question, If not symbols and rules, what else? Their model showed that one need not store cases at all; instead, a designer could tune a simulated multilayer perceptron (MLP) neural network[9] by training it to respond to specific situations and then having it respond to other situations in ways that are (the designer hopes) appropriate extrapolations of the responses it has learned. Indeed, the most striking difference between neural-network modeling and GOFAI is that the neural-network modeler provides not rules relating features of the domain but a history of training input-output pairs, and the network organizes itself by adjusting its many parameters so as to map inputs into outputs, that is, situations into responses. Thus computers running simulations of such nets do not count as physical symbol systems. Paul Smolensky, one of the PDP researchers, sums up the point:

> Connectionist systems are large networks of extremely simple processors, massively interconnected and running in parallel. Each processor has a numerical activation value which it communicates to other processors along connections of varying strengths. The activation value for each processor constantly changes in response to the activity of the processors to which it is connected. The values of some of the processors form the input to the system, and the values of other processors form the output. The connections between the processors determine how input is transformed to output. In connectionist systems, knowledge is encoded not in symbolic structures but rather in the pattern of numerical strengths of the connections between processors.[10]

In retrospect, the stages of my critique of attempts to use computers as physical symbol systems to simulate intelligence now fell into place. My early appeal to holism, my concern with commonsense understanding as know-how, Stuart's phenomenology of everyday skills, and the capacities of simulated neural networks all added up to a coherent position—one that predicted and explained why GOFAI research should degenerate just as it had.

Here is where I would like to say "and the rest is history," but there are two issues that must be faced before we lay the whole controversy to rest. First, the GOFAI research program has refused to degenerate gracefully and is fighting on, and we have to ask why this is happening. Second, the question remains whether neural networks can be intelligent or whether network researchers, like AI researchers in the 1960s, are basing their hopes on ad hoc successes that may not be generalizable.

That GOFAI was not as dead as I believed was brought home to me by public television. Readers may have seen an impressive five-part series called "The Machine That Changed the World," one episode of which was devoted to AI. In that episode my objections to symbolic AI, and specifically my conclusion that in attempting to represent common sense GOFAI had run into a problem it could not solve, was played off against the claims of a lone AI researcher, Douglas Lenat. In 1984 Lenat had shared my sense of AI's stagnation:

> By the mid-1970s, after two decades of humblingly slow progress, workers in the new field of artificial intelligence had come to a fundamental conclusion about intelligent behavior in general: it requires a tremendous amount of knowledge, which people often take for granted but which must be spoon-fed to a computer. . . . Understanding even the easiest passages in common English, for example, requires a knowledge of the context, the speaker and the world at large that is far beyond the capabilities of present-day computer programs.[11]

And by 1991 his concern was even clearer: "Most of the current AI research we've read about is currently stalled."[12] Nevertheless, he is not discouraged. He heads a research team at the Microelectronics and Computer Technology Corporation (MCC) that is in the middle of a ten-year project aimed at formalizing consensus knowledge, that is, "the millions of abstractions, models, facts, rules of thumb, representations, etc., that we all possess and that we assume everyone else does."[13]

This is not the sort of knowledge that is in an ordinary encyclopedia. Rather, it is the taken-for-granted knowledge that is used by readers in understanding an encyclopedia article and, more generally, in understanding what goes on in the world. Consensus knowledge ranges from "George Bush is President of the United States" to "George Bush wears underwear" to "When George Bush is in Washington, his left foot is also in Washington." Lenat presents himself as the only person willing to take on the commonsense-knowledge problem as a major research program instead of trying to finesse it. And he is confident that, thanks to his research, "artificial intelligence is within our grasp."[14]

Through cross-cut interviews, the Black Knight of AI, as I have been called, met the White Knight of symbolic information processing for a final joust. Lenat claimed that his project was going well and had a 60 percent chance of success. I came across as dubious but ill-informed and made some unconvincing objections. Clearly, my claim that the GOFAI program is degenerating can be dismissed as merely reporting a transient sociological phenomenon unless I can defend my conviction that Lenat's project is doomed.

To understand my critique of the GOFAI approach to common sense, it helps to know its ancestry. Rationalists such as Descartes and Leibniz thought of the mind as defined by its capacity to form representations of all domains of activity. These representations were taken to be theories of the domains in question, the idea being that representing the fixed, context-free features of a domain and the principles governing their interaction explains the domain's intelligibility. On this view all that we know—even our general know-how for getting around in the world and coping with things and people—must be mirrored in the mind in propositional form. I shall call this view of the mind and its relation to the world "representationalism." Representationalism assumes that underlying everyday understanding is a system of implicit beliefs.

This assumption is shared by intentionalist philosophers such as Edmund Husserl and computationalists such as Jerry Fodor and GOFAI researchers. The specific AI problem of representing all this knowledge in *formal* rules and features only arises after one has already assumed that common sense derives from *a vast data base of propositional knowledge*. When, instead of developing philosophical theories of the transcendental conditions that must hold if the mind is to represent the world, or proposing psychological models of how the storage and retrieval of propositional representations works, researchers in AI actually tried to formulate and organize everyday consensus knowledge, they ran into what has come to be called the commonsense-knowledge problem. There are really at least three problems grouped under this rubric:

1. How everyday knowledge must be organized so that one can make inferences from it.

2. How skills or know-how can be represented as knowing-that.

3. How relevant knowledge can be brought to bear in particular situations.

While representationalists have written programs that attempt to deal with each of these problems, there is no generally accepted solution, nor is there a proof that these problems cannot be solved. What is clear is that all attempts to solve them have run into unexpected difficulties, and this in turn suggests that there may well be in-principle limitations on representationalism. At the very least these difficulties lead us to question why anyone would expect the representationalist project to succeed.

Lenat, however, thinks that his predecessors have simply not tried hard enough to systematize common sense. His goal is to organize commonsense knowledge using general categories that make no reference to the specific uses to which the knowledge is to be put:

> Naturally, all programs are built on some primitives (predicates, frames, slots, rules, functions, scripts).[15] But if you choose task-specific primitives, you'll win in the short run (building a program for that narrow domain) but lose in the long run (you'll find yourself painted into a corner when you try to scale the program up).[16]

Lenat relates his work to the traditional philosophical job of working out an ontology—a description of the various types of context-free entities and their relationships—and he sees that turning traditional ontology into a research program is no small task:

> A serious attempt at [capturing consensus knowledge] would entail building a vast knowledge base, one that is 10^4 to 10^5 larger than today's typical expert system, which would contain general facts and heuristics and contain a wide sample of specific facts and heuristics for analogizing as well. . . . Moreover, this would include beliefs, knowledge of others' (often grouped by culture, age group, or historical era) limited awareness of what we know, various ways of representing things, knowledge of which approximations (micro-theories) are reasonable in various contexts, and so on.[17]

The data structures must represent objects and their properties, individuals, collections, space, time, causality, events and their elements, agency, institutions, and oddly, from a traditional philosophical point of view, recurrent social situations such as dinner at a restaurant or a birthday party. This data-base ontology, like any traditional rationalist ontology, must bottom out in primitive elements.

Choosing a set of representation primitives (predicates, objects, functions) has been called *ontological engineering*—that is, defining the categories and relationships of the domain. (This is empirical, experimental engineering, as contrasted with *ontological theorizing*, which philosophers have done for millennia.)[18]

Lenat is clear that his ontology must be able to represent our commonsense background knowledge—the understanding we normally take for granted. He would hold, however, that it is premature to try to give a computer the skills and feelings required for actually coping with things and people. No one in AI believes anymore that by 2001 we will have an artificial intelligence like HAL. Lenat would be satisfied if the Cyc data base could understand books and articles, for example, if it could answer questions about their content and gain knowledge from them. In fact, it is a hard problem even to make a data base that can understand simple sentences in ordinary English, since such understanding requires vast background knowledge. Lenat collects some excellent examples of the difficulty involved. Take the following sentence:

Mary saw a dog in the window. She wanted it.[19]

Lenat asks:

Does "it" refer to the dog or the window? What if we'd said "She *smashed* it," or "She pressed her nose up against it"?[20]

Note that the sentence seems to appeal to our ability to imagine how we would feel in the situation, rather than requiring us to consult *facts* about dogs and windows and how a typical human being would react. It also draws on know-how for getting around in the world, such as how to get closer to something on the other side of a barrier. In this way the feelings and bodily coping skills that were excluded to make Lenat's problem easier return. We need to be able to imagine feeling and doing things in order to organize the knowledge we need to understand typical sentences. There are also all the problems of "deixis," that is, the way we locate things with respect to our own locations, as "over there," "nearby," etc. All these problems point to the importance of the body. Lenat does not tell us how he proposes to capture in propositional terms our bodily sense of what is inside and outside, accessible and inaccessible, and what distance we need to be from various sorts of things to get an optimal grip on them. He just tells us dogmatically that this can be done.

> Our response—in principle and in CYC—is to describe perception, emotion, motion, etc., down to some level of detail that enables the system to understand humans doing those things, and/or to be able to reason simply about them.[21]

In our constructed television debate my claim that an intelligence needs a body was dismissed by reference to the case of Madeleine, a wheelchair-bound woman described by Oliver Sacks, who was blind from birth, could not use her hands to read braille, and yet acquired commonsense knowledge from books that were read to her. But this case does not in fact support Lenat. Madeleine is certainly not like a computer. She is an expert at speaking and interacting with people and so has commonsense social skills. Moreover, she has feelings, both physical and emotional, and a body that has an inside and outside and can be moved around in the world. Thus she can empathize with others and to some extent share the skillful way they encounter their world. Her expertise may well come from learning to discriminate many imagined cases and what typically occurs in them, not from forming a model of the world in Lenat's sense. Indeed, Sacks says that Madeleine had "an imagination filled and sustained, so to speak, by the images of others, images conveyed by language."[22] Thus the claim that Madeleine's acquisition of commonsense knowledge from books despite her inability to see and move her hands proves that a person can acquire and organize facts about the world on the model of a symbolic computer being fed rationally ordered representations ignores the possibility that a person's bodily skills and imagination are a necessary condition for acquiring common sense even from books.

Mark Johnson gives a good argument for the importance of imagination even in conscious problem solving:

> Imagination is a pervasive structuring activity by means of which we achieve coherent, patterned, unified representations. It is indispensable for our ability to make sense of our experience, to find it meaningful. The conclusion ought to be, therefore, that imagination is absolutely central to human rationality, that is, to our rational capacity to find connections, to draw inferences, and to solve problems.[23]

To assume that Madeleine's body and imagination are irrelevant to her accumulation, organization, and use of facts, and that her skills themselves are the result of just more storing and organizing of facts, begs the question. Why should we assume that the imagination and skills Madeleine brings to the task of learning and using common sense can be finessed by giving a computer facts and rules for organizing them?

A way to see the implausibility of this claim is to ask how the computer—with its millions of facts organized for no particular purpose—might be able to retrieve just the relevant information for understanding a sentence uttered in a specific situation. This is a far harder problem than that of answering questions on the basis of stored data, which seems to be all that Lenat has considered until now. In order to retrieve relevant facts in a specific situation, a computer would have to categorize the situation, then search through all its facts following rules for finding those that could possibly be relevant in this type of situation, and finally deduce which of these facts are actually relevant in this particular situation. This sort of search would clearly become more difficult as one added more facts and more rules to guide it. Indeed, AI researchers have long recognized that the more a system knows about a particular state of affairs, the longer it takes to retrieve the relevant information, and this presents a general problem where scaling up is concerned. Conversely, the more a human being knows about a situation or individual, the easier it is to retrieve other relevant information. This suggests that human beings use forms of storage and retrieval quite different from the symbolic one representationalist philosophers and Lenat have assumed.

Lenat admits that there is a problem:

> The natural tendency of any search program is to slow down (often combinatorially explosively) as additional assertions are added and the search space therefore grows. . . . [T]he key to preserving effective intelligence of a growing program lies in judicious adding of meta-knowledge.[24]

The problem is that the rules and meta-rules are just more meaningless facts and so may well make matters worse.

In the end, Lenat's faith that Cyc will succeed is based neither on arguments nor on actual successes but on the untested traditional assumption that human beings have a vast library of commonsense knowledge and somehow solve the scaling-up problem by applying further knowledge:

> We're often asked how we expect to efficiently "index"—find relevant partial matches—as the knowledge base grows larger and larger. . . . Our answer . . . often appears startling at first glance: wait until our programs *are* finding many, far-flung analogies, but inefficiently, i.e. only through large searches. Then investigate what additional knowledge *people* bring to bear, to eliminate large parts of the search space in those cases. Codify the knowledge so extracted, and add it to the system.[25]

But the conviction that people *are* storing context-free facts and using meta-rules to cut down the search space is precisely the dubious rationalist assumption in question. It must be tested by looking at the phenomenology of everyday know-how. Such an account is worked out by Heidegger and his followers such as Merleau-Ponty and the anthropologist Pierre Bourdieu. They find that what counts as the facts depends on our everyday skills. In describing a society in which gift-exchange is important, Bourdieu tells us:

> If it is not to constitute an insult, the counter-gift must be *deferred* and *different*, because the immediate return of an exactly identical object clearly amounts to a refusal. . . . It is all a question of style, which means in this case timing and choice of occasion, for the same act—giving, giving in return,

offering one's services, paying a visit, etc.—can have completely different meanings at different times.[26]

Yet members of the culture have no trouble understanding what to do. Once one has acquired the necessary social skill, one does not need to *recognize* the situation objectively as having the features of one in which gift-giving is appropriate and then *decide* rationally what gift to give. Normally one simply responds in the appropriate circumstances by giving an appropriate gift. That this is the normal response is what constitutes the circumstance as a gift-giving situation. The same, of course, holds for the know-how of what gift is appropriate. One does not have to *figure out* what is appropriate, or at least not the range of what is appropriate. Everyone's skills are coordinated so that normally one is just solicited by the situation to give a certain type of gift, and the recipient, socialized into the same shared practices, finds it appropriate. Bourdieu comments:

> The active presence of past experiences . . . deposited in each organism in the form of schemes of perception, thought, and action, tend to guarantee the 'correctness' of practices and their constancy over time, more reliably than all formal rules and explicit norms.[27]

This sort of experience suggests that structuring a knowledge base so as to represent all facts about gift-giving is necessary only for a stranger or spectator who does not already have the appropriate skill. Bourdieu insists that it is a mistake—one often made by anthropologists, philosophers, and, we can add, AI researchers—to read the rules we need to appeal to in breakdown cases back into the normal situation and then to appeal to such representations for a causal explanation of how a skillful response is normally produced.

The point of this example is that knowing how to give an appropriate gift at the appropriate time and in the appropriate way requires cultural *savoir faire*. So knowing what a gift is is not a bit of factual knowledge, separate from the skill or know-how for giving one. The distinction between what a gift is and what counts as a gift, which seems to distinguish facts from skills, is an illusion fostered by the philosophical belief in a nonpragmatic ontology. Since the organization and content of a gift frame presupposes gift-giving *practices*, there is no need of and no evidence for frames and slots that spell out the supposed objective features of gifts and gift-giving occasions. It is doubtful that such an exhaustive account is even possible.

There is an even deeper problem lurking in wait for Lenat. We are not only able to cope with changing events and motivations, both on the fly and in our imaginations; we are also able to project our understanding into new situations. If one is a master of a cultural practice, one can sometimes do what has not so far counted as appropriate and have it recognized in retrospect as having been just the right thing to do. Thus a master of the culture can introduce a new sort of gift or a new gift-giving occasion. This happens not only at private parties and public ceremonies but, of course, also in stories and reports about such occasions. A data base that did not share the culture's *savoir faire*, then, would not only be unable to *give* appropriate gifts—we've agreed that Lenat does not have to build a skillful robot—but would also fail as a system storing our consensus knowledge of gifts because it could not *understand* innovations in gift-giving.

Lenat sees that the ability to extend our knowledge is crucial both to exercising our knowledge and to acquiring new knowledge from experience. Indeed, the success of his program, by his own criterion, requires that it be able to extend what it knows and learn from reading books and articles rather than from "brain surgery." The question of how we project or extend what we already know therefore becomes crucial.

> Designing more proficient learning programs depends in part on finding ways to tap a source of power at the heart of human intelligence: the ability to understand and reason by analogy. . . . This source of power is only beginning to be exploited by intelligent software, but it will doubtless be the focus of future research.[28]

Granted that an intelligent person can see analogies or similarities to what he or she already knows, there are several ways to think about this basic human capacity. The classical rationalist tradition since Aristotle has tried to understand analogies as proportions. A second tradition traces analogy back to our experience of our body. A third approach has reacted to the implausibility of the classical tradition by approaching analogy in terms of extrapolating a style. Lenat, of course, belongs to the first camp; Mark Johnson and George Lakoff belong to the second; and Heidegger, Merleau-Ponty, and Bourdieu belong to the third.

Bourdieu mentions the role of style in learning and extrapolating gift exchanges. It may well be that different cultures have different styles—aggressive, passive, controlling, etc.—and that infants pick up first on these pervasive styles, which in turn direct what the infant notices and imitates. If this is so, we would expect style to play an important role in what similarities are noticed and what metaphors are used to organize experience. Style, however, is generally neglected in GOFAI. The only AI researcher who has seen its importance is Douglas Hofstadter, but even he has not come up with any convincing proposals for dealing with it.[29]

Given that the solution of this problem is essential to the success of his system, Lenat's remarks on analogy are rather sketchy. Still, what he does say works out the rationalist approach in enough details to reveal its implausibility. Lenat's sensitivity to metaphors and analogies is arresting and persuasive.

> Almost every sentence is packed with metaphors and analogies. An unbiased example: here is the first article we saw today (April 7, 1987), the lead story in the *Wall Street Journal*: "Texaco lost a major ruling in its legal battle with Pennzoil. The Supreme Court dismantled Texaco's protection against having to post a crippling $12 billion appeals bond, pushing Texaco to the brink of a Chapter 11 filing." Lost? Major? Battle? Dismantled? Posting? Crippling? Pushing? Brink? The example drives home the point that, far from overinflating the need for real-world knowledge in language understanding, the usual arguments about disambiguation barely scratch the surface. (Drive? Home? The point? Far? Overinflating? Scratch? Surface? Oh no, I can't call a halt to this! (call? halt?))[30]

Lenat's faith that the metaphors "bottom out" in primitives, however, is not convincing, nor is it argued for. He simply asserts:

> These layers of analogy and metaphor eventually 'bottom out' at physical—*somatic*—primitives: up, down, forward, back, pain, cold, inside, seeing,

sleeping, tasting, growing, containing, moving, making noise, hearing, birth, death, strain, exhaustion, . . .[31]

The fact that these are *somatic* primitives does not seem to bother him at all.

Lenat, nonetheless, asks the right question: "How can a program automatically find good mappings?"[32] But he gives the simplistic rationalist answer: "If *A* and *B* appear to have some unexplained similarities, then it's worth your time to hunt for additional shared properties."[33]

This begs the question. Everything is similar to everything else in an indefinitely large number of ways. Why should we suppose that any two items should be compared? Even if two frames have many slots in common, why should we think these are the important similarities? Perhaps the important similarities cannot be symbolically represented at all. Both the defenders of the basic role of our sense of our active body with inside/outside, forward/backward, and up/down dimensions and those who hold that similarity of style is what defines what is worth comparing would hold that there is no reason to think that the constraints on similarity can be represented symbolically.

When John Searle tried to understand metaphors as proportions, he found that metaphors like "Sally is a block of ice" could not be analyzed by listing the features that Sally and a large, cold cube have in common.

If we were to enumerate quite literally the various distinctive qualities of blocks of ice, none of them would be true of Sally. Even if we were to throw in the various beliefs that people have about blocks of ice, they still would not be literally true of Sally. . . . Being unemotional is not a feature of blocks of ice because blocks of ice are not in that line of business at all, and if one wants to insist that blocks of ice are literally unresponsive, then we need only point out that that feature is still insufficient to explain the metaphorical utterance meaning . . . because in that sense bonfires are "unresponsive" as well.[34]

Searle concludes:

There are . . . whole classes of metaphors that function without any underlying principles of similarity. It just seems to be a fact about our mental capacities that we are able to interpret certain sorts of metaphor without the application of any underlying 'rules' or 'principles' other than the sheer ability to make certain associations. I don't know any better way to describe these abilities than to say that they are nonrepresentational mental capacities.[35]

So far we have only discussed the facts and metaphors that are constituted by our *social* skills. What about the facts of nature? Where a domain of facts is independent of us, as is the domain of physical objects, do we then need a theory of the domain? Not likely. The way people cope with things is sometimes called commonsense physics. This leads to the comforting illusion that just as the planets do not move around at random but obey general principles, so everyday objects do not stick, slide, fall, and bounce in an unprincipled way but obey complex and particular laws. Attempts to work out commonsense physics for the simplest everyday objects, however, lead to formal principles that are subject to many exceptions and are so complex that it is hard to believe they could be in a child's mind.[36] Lenat concludes from this that what we know concerning how everyday objects behave cannot be principles but must be a lot of facts and rules.

[The Cyc] methodology will collect, e.g., all the facts and heuristics about "Water" that newspaper articles assume their readers already know. This is in contrast to, for instance, naive physics and other approaches that aim to somehow capture a deeper theory of "Water" in all its various forms.[37]

But granted that there is no reason to think that there can be a *theory* of commonsense physics as there is of celestial physics, that is no reason to think that our know-how for dealing with physical objects can be spelled out in some all-purpose data base concerning physical objects and their properties. Perhaps there is no set of context-free facts adequate to capture the way everyday things such as water behave. We may just have to learn from vast experience how to respond to thousands of typical cases. That would explain why children find it fascinating to play with blocks and water day after day for years. They are probably learning to discriminate the sorts of typical situations they will have to cope with in their everyday activities. For natural kinds like water, then, as well as for social kinds like gifts, common sense seems to be based on knowing-how rather than knowing-that, and this know-how may well be a way of storing our experience of the world that does not involve representing the world as symbolic AI required.

This still leaves the important question of how human beings manage to engage in purposive behavior. The traditional view, accepted by GOFAI, has been that they use their theory of the domain in question to work out a plan for accomplishing whatever they are trying to do. But rather than suggesting that people store vast numbers of facts and then plan how to use them, the phenomena, which have to be trusted until psychology or neuroscience gives us any reason to think otherwise, suggest that when one has had a great deal of experience in a domain, one simply sees what needs to be done. It seems that when a person has enough experience to make him or her an expert in any domain, the field of experience becomes structured so that one directly experiences which events and things are relevant and how they are relevant. Heidegger, Merleau-Ponty, and the gestaltists would say that objects appear to an involved participant not in isolation and with context-free properties but as things that solicit responses by their significance.

In the first edition of this book I noted that good chess players don't seem to figure out from scratch what to do each time they make a move. Instead, they zero in on a certain aspect of the current position and figure out what to do from there (pp. 102–106). In *Mind Over Machine* Stuart went further and pointed out that a mere master might need to figure out what to do, but a grandmaster just sees the board as demanding a certain move.[38]

We are all masters in our everyday world. Consider the experience of entering a familiar type of room. We know but do not appeal to the sort of facts that can be included in a room frame, such as that rooms have floors, ceilings, and walls, that walls can have windows in them, and that the floor can have furniture on it. Instead, our feeling for how rooms normally behave, a skill for dealing with them that we have developed by crawling and walking around many rooms, gives us a sense of relevance. We are skilled at not coping with the dust, unless we are janitors, and not paying attention to whether the windows are open or closed, unless it is hot, in which case we know how to do what is appropriate. Our expertise in dealing with rooms determines from moment to moment both what we cope with by using and what we cope with by ignoring (while being ready to use it should an appropriate occasion arise). This global familiarity maps our past experience of the room onto our current activity, so that what is appropriate on each occasion is experienced as perceptually salient or simply elicits what needs to be done.

In general, human beings who have had vast experience in the natural and social world have a direct sense of how things are done and what to expect. Our global familiarity thus enables us to respond to what is relevant and ignore what is irrelevant without planning based on purpose-free representations of context-free facts. Such familiarity differs entirely from our knowledge of an unfamiliar room, such as the room of an seventeenth-century nobleman. In that sort of room our knowledge resembles the sort of knowledge a data base might have. But even if a Jacobean drawing-room frame and its slots were all in place, we would still be disoriented. We would not know what to pay attention to or how to act appropriately.

Global sensibilities (or the imagination thereof) determine situational relevance because our world is organized by these preconceptual meanings. It is in terms of them that objects and events are experienced *as* something. Our everyday coping skills and the global familiarity they produce determine what counts as the facts and the relevance of all facts and so are already presupposed in the organization of the frames and slots GOFAI uses for representing these facts. That is why human beings cope more easily and expertly as they learn to discriminate more aspects of a situation, whereas, for data bases of frames and rules, retrieving what is relevant becomes more and more difficult the more they are told.

Lenat does seem to be correct in seeing the Cyc project as the last defense of the AI dream of producing broad, flexible human intelligence. Indeed, just because of its courage and ambition, the Cyc project, more than any previous one, confronts the problems raised by the idea of basing intelligence on symbolic representations. As we have just seen, the somatic and stylistic background sensitivities that determine what counts as similar to what and the background coping familiarity that determines what shows up as relevant are presupposed for the intelligent use of the facts and rules with which symbolic AI starts. The hope that these background conditions can be analyzed in terms of the features whose isolation and recognition they make possible is, on the face of it, implausible. The only arguments that are ever given in support of the physical symbol system hypothesis are the rationalist assumption that understanding equals analysis, so that all of experience must be analyzable (that is, there must be a theory of every intelligible domain), or the GOFAI response that the mind must be a symbol manipulator since no one knows what else it might be. Now that both of these arguments have lost plausibility, there remains only the pragmatic argument that GOFAI will demonstrate its possibility by producing an intelligent machine. So far that sort of claim has not been made good, and Cyc faces all the old problems in their most daunting form. The project has five more years to go, but Lenat has given us no reason to be optimistic. It seems highly likely that the rationalist dream of representationalist AI will be over by the end of the century.

For three groups of AI researchers whose work now focuses on alternative approaches, GOFAI is already over. One of these approaches, associated with the work of Philip Agre and David Chapman, attempts to produce programs that interact intelligently with a microworld without using either context-free symbolic representations or internal model-based planning. The second, represented by the neural-network modelers, abandons representation altogether. This approach uses conventional features but produces outputs by a direct mapping from the inputs, with the mapping extrapolated from examples provided by an expert. A third new approach to AI, called reinforcement learning, aims to develop a program that dispenses with the expert and uses actual performance in the skill domain in order to find, on its own, a successful input-output rule. It is worth considering the advantages and limitations of each of these approaches.

The interactionists are sensitive to the Heideggerian critique of the use of symbolic models of the world and attempt to turn Heidegger's account of ongoing skillful coping[39] into an alternative research program. At MIT, where this approach was developed, it is sometimes called Heideggerian AI. Terry Winograd, who was the first to introduce Heidegger into his computer science courses, has described this surprising new development:

For those who have followed the history of artificial intelligence, it is ironic that [the MIT] laboratory should become a cradle of "Heideggerian AI." It was at MIT that Dreyfus first formulated his critique, and, for twenty years, the intellectual atmosphere in the AI Lab was overtly hostile to recognizing the implications of what he said. Nevertheless, some of the work now being done at that laboratory seems to have been affected by Heidegger and Dreyfus.[40]

The AI Lab work Winograd is referring to is the influential theory of activity developed by Agre and Chapman, implemented in two programs, Pengi and Sonja, that play computer games. Agre and Chapman question the need for an internal symbolic model of the world that represents the context-free features of the skill domain. Following Heidegger, they note that in our everyday coping we experience ourselves not as subjects with mental representation over against objects with fixed properties, but rather as absorbed in our current situation, responding directly to its demands.

Interactive AI takes seriously the view I attributed to Heidegger in this book—that there is usually no need for a representation of the world in our mind since the best way to find out the current state of affairs is to look to the world as we experience it. Chapman tells us:

If you want to find out something about the world that will affect how you should act, you can usually just look and see. Concrete activity is principally concerned with the here-and-now. You mostly don't need to worry about things that have gone before, are much in the future, or are not physically present. You don't need to maintain a world model; the world is its own best representation.[41]

Agre and Chapman also adapt another Heideggerian thesis that Stuart and I developed in *Mind Over Machine*, namely that behavior can be purposive without the agent having in mind a goal or purpose.

In a great many situations, it's obvious what to do next given the configuration of materials at hand. And once you've done that the next thing to do is likely to be obvious too. Complex sequences of actions result, without needing a complex control structure to decide for you what to do.[42]

What is original and important in Agre and Chapman's work is that these ideas are taken out of the realm of armchair phenomenology and made specific enough to be implemented in programs. What results is a system that represents the world not as a set of objects with properties but as current functions (what Heidegger called in-order-tos). Thus, to take a Heideggerian example, I experience a hammer I am using not as an object with properties but as in-order-to-drive-in-the-nail. Only if there is some disturbance does the skilled performer notice what I have called aspects of the situation. In Heidegger's example, the carpenter notices that the hammer is too heavy. Both of the above ways of being, which Heidegger calls the available (the ready-to-hand) and the unavailable (the unready-to-hand), are to be distinguished from what he calls the occurrent (the present-at-hand) mode of being, the mode of being of stable objects. Objects can be recognized as the same even when they are used in different contexts or when some of their properties change. Such reidentifiable objects with their changing features or properties have been the only mode of being represented in GOFAI models. The interactionists seek to represent the available and the unavailable modes. Chapman speaks in this respect of "deictic representations":

> The sorts of representations we are used to are *objective*: they represent the world without reference to the representing agent. *Deictic* representations represent things in terms of their relationship with the agent. The units of deictic representation are *entities*, which are things in a particular relationship to the agent, and relational *aspects* of these entities. For example, *the-cup-I-am-drinking-from* is the name of an entity, and *the-cup-I-am-drinking-from-is-almost-empty* is the name of an aspect of it. *The-cup-I-am-drinking-from* is defined in terms of an agent and the time the aspect is used. The same representation refers to different cups depending on whose representation it is and when it is used. It is defined *functionally*, in terms of the agent's purpose: drinking.[43]

The other important Heidegger-inspired innovation in interactive programming is its implementation of purposive action. A GOFAI planner searches the space of possible sequences of actions to determine how to get from a symbolic representation of the current situation to a specified goal. The interactive approach to action stipulates a mapping from situations directly to actions.

Interactive AI has implemented Heidegger's phenomenology of everyday coping but has not attempted to implement his account of the background familiarity on the basis of which certain equipment is seen as relevant and certain courses of action solicit my response. This gap shows up in Chapman's unsatisfying account of relevance. Chapman tells us that "agents represent only relevant aspects of the situation."[44] But this turns out to mean that, as in all GOFAI programs, the programmer has predigested the domain and determined for the system what are the possibly relevant features at any given moment.

So far it looks like Heideggerian AI is true to Heidegger's phenomenology in what it leaves out—long-range planning and internal representations of reidentifiable objects with context-free features—but it lacks what any intelligent system needs, namely the ability to discriminate relevant distinctions in the skill domain and to learn new distinctions from experience. To provide this crucial capability, more and more researchers are looking to simulated neural networks. We therefore turn to the question of whether such networks can exhibit what I have called familiarity or global sensitivity and, if not, whether they can cope in some other way with relevance and learning. (My use of "we" here is not royal but literal, since my brother Stuart has made indispensable contributions to the rest of this introduction.)

We have already mentioned that neural-network modeling, the fashionable answer to the what-else-could-it-be question, has swept away GOFAI and given AI researchers an optimism they have not had since the 1960s. After all, neural networks can learn to recognize patterns and pick out similar cases, and they can do this all in parallel, thus avoiding the bottleneck of serial processing. But neural networks raise deep philosophical questions. It seems that they undermine the fundamental rationalist assumption that one must have abstracted a theory of a domain in order to behave intelligently in that domain. In its simplest terms, as understood from Descartes to early Wittgenstein, finding a theory means finding the invariant features in terms of which one can map specific situations onto appropriate responses. In physical symbol systems the symbols in the representation are supposed to correspond to these features, and the program maps the features onto the response. As we saw, Lenat, the last heir to GOFAI, assumes that there must be such context-free primitives in which his ontology would bottom out. When neural networks became fashionable, traditional AI researchers assumed that the hidden nodes in a trained net would detect and learn the relevant features, relieving the programmer of the need to discover them by trial and error. But this turned out to be problematic.

The input to neural networks must, of course, be expressed in terms of stable, recognizable features of the domain. For example, a network that is to be trained to play chess would take as its inputs board positions defined in terms of types and locations of pieces. The question is whether a network that has learned to play chess has detected higher-order features, such as unbalanced pawn structure, that combine these input features in such a way that any position that shares the same higher-order features maps into the same move. If a given network architecture trained on a given set of examples could be shown to detect such higher-order features independently of its connection strengths prior to training, then it could be said to have abstracted the theory of the domain. If, for example, such features turned out to be the kinds of features chess masters actually think about, then the net would have discovered the theory of the chess domain that chess theorists and symbolic AI researchers have sought for so long. If these higher-order features were not the sort of features an expert in the domain could recognize, the belief that programmers of AI systems could invent higher-order features based on chess knowledge would of course be shaken, but the assumption that there must be a theory of any domain in which intelligent behavior is possible would not have been called into question.

The implications for rationalism, however, may be much more serious. To defend the theory theory, rationalists might well insist that, given any particular set of connection strengths as a starting point for training a network with examples, we can always identify higher-order features, even if these features cannot be used consciously by experts. Consider the simple case of layers of binary units activated by feedforward, but not lateral or feedback, connections. To construct such higher-order features from a network that has learned certain associations, we could interpret each node one level above the input nodes, on the basis of the connections to it, as detecting when any one of a certain set of identifiable input patterns is present. (Some of the patterns will be the ones used in training, and some will never have been used.) If the set of input patterns that a particular node detects is given a name (it almost certainly won't have one already), the node could be interpreted as detecting the highly abstract feature so named. Hence, every node one level above the input level could be characterized as a feature detector. Similarly, every node a level above those nodes could be interpreted as detecting a higher-order feature defined as the presence of one of a specified set of patterns among the first level of feature detectors. And so on up the hierarchy. A similar story could be constructed for neurons with graded (continuous, nonbinary) responses. One would then speak of *the extent to which* a higher-order feature is present.

The fact that intelligence, defined as the knowledge of a certain set of associations appropriate to a domain, can always be accounted for in terms of relations among a number of such highly abstract features of a skill domain does not, however, preserve the rationalist intuition that these explanatory features capture the essential structure of the domain. The critical question is whether, if several different nets with different initial connection strengths were trained to produce a given set of input/output mappings, the same higher-order features would be detectable in all of them or, at least, whether, at some level of abstraction, all of the nets could be seen as abstracting equivalent invariances.

No such invariances have been found. The most thorough search concerns a neural network called NETtalk that converts printed text into speech. NETtalk is given several pages of text plus the correct pronunciation of the middle letter of every string of seven characters in the text. The net starts with random connection strengths, and its reading of the text sounds like noise. After many hours of training using backpropagation, a technique that changes the connection strengths repeatedly, each time bringing the actual output closer to the correct output, the net learns to read the text aloud in a way that a native speaker can easily understand.[45] But careful analysis of the activity of the hidden nodes when the net was producing correct responses failed to reveal any consistent higher-order features in trials with different initial connection strengths. Thus we can say that so far neural-network research has tended to substantiate the belief that coping does not require the abstraction of a theory of the skill domain.[46] This is bad news for rationalism but gives networks a great advantage over GOFAI.

Nevertheless, the commonsense-knowledge problem resurfaces in this work and threatens its progress just as it did work in GOFAI. All multilayer perceptron neural-network modelers agree that an intelligent network must be able to generalize; for example, for a given classification task, given sufficient examples of inputs associated with one particular output, it should associate further inputs of the same type with that same output. But what counts as the same type? The network's designer usually has in mind a specific definition of "type" required for a reasonable generalization and counts it a success if the net generalizes to other instances of this type. But when the net produces an unexpected association, can one say that it has failed to generalize? One could equally well say that the net has all along been acting on a different definition of "type" and that that difference has just been revealed.

For an amusing and dramatic case of creative but unintelligent generalization, consider one of connectionism's first applications. In the early days of this work the army tried to train an artificial neural network to recognize tanks in a forest. They took a number of pictures of a forest without tanks and then, on a later day, with tanks clearly sticking out from behind trees, and they trained a net to discriminate the two classes of pictures. The results were impressive, and the army was even more impressed when it turned out that the net could generalize its knowledge to pictures that had not been part of the training set. Just to make sure that the net was indeed recognizing partially hidden tanks, however, the researchers took more pictures in the same forest and showed them to the trained net. They were depressed to find that the net failed to discriminate between the new pictures of trees with tanks behind them and the new pictures of just plain trees. After some agonizing, the mystery was finally solved when someone noticed that the original pictures of the forest without tanks were taken on a cloudy day and those with tanks were taken on a sunny day. The net had apparently learned to recognize and generalize the difference between a forest with and without shadows! This example illustrates the general point that a network must share our commonsense understanding of the world if it is to share our sense of appropriate generalization.

One might still hope that networks different from our brain will make exciting new generalizations and add to our intelligence. After all, detecting shadows is just as legitimate as detecting tanks. In general, though, a device that could not learn our generalizations and project our practices to new situations would just be labeled stupid. For example, thanks to our bodies, we normally see symmetric objects as similar. If a system consistently classified mirror images of otherwise identical objects as different but classified objects that cast the same shadows or had any red on them as similar, we would count it not as adding to our intelligence but as being unteachable or, in short, stupid as far as joining our community or giving us new insights was concerned. For an exercise in interesting but unintelligible categorization, consider Jorge Luis Borges's story of "a 'certain Chinese encyclopedia' in which it is written that 'animals are divided into: (a) belonging to the Emperor, (b) embalmed, (c) tame, (d) sucking pigs, (e) sirens, (f) fabulous, (g) stray dogs, (h) included in the present classification, (i) frenzied, (j) innumerable, (k) drawn with very fine camelhair brush, (l) et cetera, (m) having broken the water pitcher, (n) that from a long way off look like flies.'"[47]

Neural-network modelers were initially pleased that their nets were a blank slate (*tabula rasa*) until trained, so that the designer did not need to identify and provide anything resembling a pretraining intelligence. Recently, however, they have been forced by the problem of producing appropriate, human-like generalizations to the recognition that, unless the class of possible generalizations is restricted in an appropriate a priori manner, nothing resembling human generalizations can be confidently expected.[48] Consequently, after identifying in advance the class of allowable human-like generalizations appropriate to the problem (the hypothesis space), these modelers then attempt to design the architecture of their networks so that they transform inputs into outputs only in ways that are in the hypothesis space. Generalization would then be possible only on the designer's terms. While a few examples will be insufficient to identify uniquely the appropriate member of the hypothesis space, after enough examples only one hypothesis will account for all the examples. The network will then have learned the appropriate generalization principle. That is, all further input will produce what, from the designer's point of view, is the right output.

The problem here is that the designer has determined, by means of the architecture of the network, that certain possible generalizations will never be found. All this is well and good for toy problems in which there is no question of what constitutes a reasonable generalization, but in real-world situations a large part of human intelligence consists in generalizing in ways that are appropriate to a context. If the designer restricts the network to a predefined class of appropriate responses, the network will be exhibiting the intelligence built into it by the designer for that context but will not have the common sense that would enable it to adapt to other contexts as a truly human intelligence would.

Perhaps a network must share size, architecture, and initial-connection configuration with the human brain if it is to share our sense of appropriate generalization. Indeed, neural-network researchers with their occasional ad hoc success but no principled way to generalize seem to be at the stage of GOFAI researchers when I wrote about them in the 1960s. It looks likely that the neglected and then revived connectionist approach is merely getting its deserved chance to fail.

To generalize in the way that human beings do, a network's architecture would have to be designed in such a way that the net would respond to situations in terms of what are for human beings relevant features. These features would have to be based on what past experience has shown to be important and also on recent experiences that determine the perspective from which the situation is viewed. Only then could the network enter situations with perspective-based human-like expectations that would allow recognition of unexpected inputs (such as tanks in forests) as well as significant expected inputs that are not currently present in the situation. No current networks show any of these abilities, and no one at present knows or even speculates about how our brain's architecture produces them.

There is yet another fundamental problem with the route to artificial intelligence through the supervised training of neural networks. In GOFAI it has long been clear that whatever intelligence the system exhibits has been explicitly identified and programmed by the system designer. The system has no independent learning ability that allows it to recognize situations in which the rules it has been taught are inappropriate and to construct new rules. Neural networks do appear to have learning ability; but in situations of supervised learning, it is really the person who decides which cases are good examples who is furnishing the intelligence. What the network learns is merely how to capture this intelligence in terms of connection strengths. Networks, like GOFAI systems, therefore lack the ability to recognize situations in which what they have learned is inappropriate; instead, it is up to the human user to recognize failures and either modify the outputs of situations the net has already been trained on or provide new cases that will lead to appropriate modifications in behavior. The most difficult situation arises when the environment in which the network is being used undergoes a structural change. Consider, for example, the situation that occurred when OPEC instigated the energy crisis in 1973. In such a situation, it may well happen that even the human trainer does not know the responses that are now correct and that should be used in retraining the net. Viewed from this perspective, neural networks are almost as dependent upon human intelligence as are GOFAI systems, and their vaunted learning ability is almost illusory. What we really need is a system that learns on its own how to cope with the environment and modifies its own responses as the environment changes.

To satisfy this need, recent research has turned to an approach sometimes called "reinforcement learning."[49] This approach has two advantages over supervised learning. First, supervised learning requires that the device be told the correct action for each situation. Reinforcement learning assumes only that the world provides a reinforcement signal measuring the immediate cost or benefit of an action. It then seeks to minimize or maximize the total reinforcement it receives while solving any problem. In this way, it gradually learns from experience the optimal actions to take in various situations so as to achieve long-term objectives. To learn skillful coping, then, the device needs no omniscient teacher, just feedback from the world. Second, in supervised learning, any change in the skill environment requires new supervision by an expert who knows what to do in the new environment. In reinforcement learning, new conditions automatically lead to changes in reinforcement that cause the device to adapt appropriately.

An example will clarify what reinforcement learning in its most elemental form is all about. Suppose a device is to learn from repeated experience the shortest path from point A to point B in a city. The device knows where it is (its current state) and the possible directions it can go in (its space of allowable current actions). After it chooses an action (a direction), it observes the distance to the next intersection (its next decision point). This cost is its immediate reinforcement. It also observes the location of the next intersection (its new situation). The standard AI approach would be to have the device create an internal map of the city based on its experiences and then use that map and some computational algorithm to determine the shortest path. The new approach, like Heideggerian AI, dispenses with models and long-range planning. Instead the device repeatedly takes various paths from A to B, learning in which direction it should go at each intersection to create the shortest path from a given starting intersection to B. It does this not by trying alternative paths and remembering the best but by gradually learning only one piece of information besides its best decision at each intersection, namely the shortest distance from that intersection to B. This is the "value" of the intersection. After each decision and observation of the distance to the next intersection, the reinforcement algorithm evaluates that decision in terms of its current estimates of the value of the intersection it is at and the one to which it is going next. If it looks to be a good decision, it renders that decision more likely to be chosen in the future when the path problem is repeated and it finds itself at the same intersection. It also updates its estimate of the value of the current intersection.[50]

We have so far described a problem in which a given action in a given situation always leads to the same next situation and the same immediate reinforcement, but the approach is equally appropriate to probabilistic environments in which the device seeks actions that minimize or maximize expected long-term reinforcement. Values learned for situations are then minimal or maximal expected values. To cite one example, reinforcement-learning ideas (together with other mechanisms that are less like what brains seem to do but that speed up the learning) have been tried on the stochastic game of backgammon.[51] A program that played hundreds of thousands of games against itself, without expert-specified principles of the game or expert-supplied positional values or correct moves, learned expected values of positions well enough to play at the level of a good human tournament player. It was as proficient as any computer program using more conventional AI or supervised learning methods.[52]

All of this fits well with the phenomena. Most of our skills involve action in evolving situations and are learned from trial-and-error experience with environmental feedback but without teachers (or, sometimes, from experience-based fine-tuning of what we initially learned through instruction). Moreover, while experts generally cannot access any information explaining their ability, they can usually assess the value or desirability of a situation easily and rapidly and recommend an appropriate action.

Assuming that reinforcement-learning ideas correctly capture some of the essence of the human intelligence involved in learning skillful coping, the question naturally arises, Can one build a device that does as well as expert human beings using the phenomenologically plausible minimal essence of reinforcement learning, at least in particular skill domains? At least two improvements on present practice, neither of which appears achievable based on current knowledge, are needed. First, should reinforcement learning be applied to a problem in which the number of situations that might be encountered far exceeds the number that are actually encountered during training, some method of assigning fairly accurate actions and values to the novel situations is needed. Second, if reinforcement learning is to produce something resembling human intelligence, the reinforcement-learning device must exhibit global sensitivity by encountering situations under a perspective and by actively seeking relevant input.

Consider first the problem of behavior in unique situations. This problem has been dealt with by two procedures. The first is an automatic generalization procedure that produces actions or values in previously infrequently encountered situations on the basis of actions or values learned for other situations.[53] The second is to base one's actions on only a relevant subset of the totality of features of a situation and to attach a value to the situation based only on those relevant features; in this way, we lump together experiences with all situations sharing the same relevant features regardless of the nonrelevant ones. Actions are chosen or values learned based on experiences with situations sharing these relevant features. Both of these approaches are unsatisfactory. Concerning an automatic generalization procedure, at the point where generalization is required, the situation is identical with the one faced by supervised learning. No one has any idea how to get a network or any other mechanism to generalize in the way that would be required for human-like intelligence.

The second problem mentioned above—learning what features of a situation should be treated as a relevant subset and used in determining actions and values—is equally difficult. One can find out which features of the current state of affairs are relevant only by determining what sort of situation this state of affairs is. But that requires retrieving relevant past situations. This problem might be called *the circularity of relevance*. To appreciate its implications, imagine that the owner of a baseball team gives the team manager a computer loaded with facts about each player's performance under various conditions. One day, after consulting the computer late in the last inning, the manager decides to replace the current batter, A, with a pinch hitter, B. The pinch hitter hits a home run, and the team wins the game. The owner, however, is upset and accuses the manager of misusing the computer, since it clearly shows that B has a lower batting average than A. But, says the manager, the computer also showed that B has a higher batting average in day games, and this was a day game. Yes, responds the owner, but it also showed that he has a lower average against left-handed pitchers, and there was a leftie on the mound today. And so on. The point is that a manager's expertise, and expertise in general, consists in being able to respond to the relevant facts. A computer can help by supplying more facts than the manager could possibly remember, but only experience enables the manager to see the current state of affairs as a specific situation and so see what is relevant. That expert know-how cannot be put into the computer by adding more facts, since the issue is which is the current correct perspective from which to determine which facts are relevant.

Current procedures attempt to learn about relevance by keeping track of certain statistics during trial-and-error learning. A procedure proposed by Chapman and Kaelbling[54] starts by assuming that no features are relevant to action or value assessment, that is, that the same action should be taken no matter what the situation and that the same value should be attached to all situations. Then, for each possibly relevant feature of a situation, the procedure keeps track of statistics on how things work when that feature takes on each of its possible values (often just "present" or "not present"). If, on the basis of current statistics, the value of the feature seems to affect actions or values significantly, it is declared relevant. The situation receives an ever finer description as the set of features discovered to be relevant grows.

Something vaguely of this sort is probably what the brain does. There are, however, serious problems with the particular procedure described above and variations on it. First, a feature may not be relevant to behavior on its own but may be relevant when combined with one or more other features. To remedy this, we would need to gather statistics on the relevance of combinations of features, leading to an exponential explosion of possibly important statistics.

Second, this approach assumes that the relevance of a feature is a property of the domain; what is measured is the feature's relevance in all situations encountered. But a feature may be relevant in certain situations and not in others. We would therefore need to gather relevance data separately for each situation, again leading to exponential growth in the quantity of statistics gathered. Statistics gathering, therefore, does not seem a practical way for current computer procedures to deal with the relevance-determination aspect of intelligent behavior. As we shall see, given the size and structure of the brain, it may well be no accident that no one currently has any idea how to deal with this problem without gathering an impractical amount of statistical data.

A related third problem is that there is no limit to the number of features that might conceivably be relevant in some situations. We cannot simply start with all features that might possibly be relevant, gather statistics on each, and then leave out those that experience indicates can safely be ignored. But if we start with a finite set of possibly relevant features, there is no known way of adding new features should the current set prove inadequate to account for the learned facts about reinforcement and situation transition.

So how does the brain do it? No one knows. But certain facts seem relevant. First, it appears that experience statistically determines individual neural synaptic connections, so that the brain, with its hundreds of thousands of billions of adjustable synapses, can indeed accumulate statistical information on a scale far beyond current or foreseeable computers. Second, the reinforcement-learning procedures now being studied generally produce simple stimulus-response behavior in the sense that the input, a situation description, maps directly forward into the output, an action or situation value. The brain clearly has internal states that we experience as moods, anticipations, and familiarities that are correlated with the current activity of its hidden neurons when the input arrives. These are determined by its recent inputs as well as by the synaptic connection strengths developed on the basis of long-past experiences, and these as well as the input determine the output. One can in principle include such internal states in reinforcement-learning procedures by adding the current internal state of the device to the situation description, and a few researchers have moved in this direction. In effect, such an extended procedure in which the internal state is viewed as the perspective brought to the problem based on recent events would allow the incorporation of perspective into neural models. But since no one knows how to incorporate internal states appropriately, a breakthrough will be necessary before human behavior can be imitated successfully.

Most important, there is evidence that the internal brain state interacts with an input and then feeds its output to motor-control neurons as well as back into the input pathways, affecting receptors through motor control so that they actively seek information and simultaneously influencing perceived relevance through the feedback into input pathways. This would be the brain basis of the phenomenon of global sensitivity that enables a skilled person to see directly what is relevant in his or her skill domain. This feedback based on the interaction of sensory input and internal brain state would be a powerful mechanism for dealing with information pickup and relevance problems, but currently no details of this mechanism are understood or even hypothesized in a way that could guide AI research. It thus seems reasonable to hold that mechanisms exist in the brain that can in principle be understood and duplicated in hardware so as to produce artificial intelligence in restricted domains and that reinforcement learning is a small step in the right direction, while simultaneously holding that our current ignorance concerning the brain and practical limitations on computer memory size make it highly unlikely that there will be substantial progress toward this kind of brain-inspired AI in the foreseeable future.

One problem would remain even if the above practical problems were solved. In all applications of reinforcement learning the programmer must use his or her knowledge of the problem to formulate a rule that specifies the immediate reinforcement received at each step. For path problems and games the objective nature of the problem dictates the rule. If, however, the problem involves human coping, there is no simple objective answer as to what constitutes immediate reinforcement. Even if we assume the simplistic view that human beings behave so as to maximize their total sense of satisfaction, a reinforcement-learning approach to producing such behavior would require a rule for determining the immediate satisfaction derived from each possible action in each possible situation. But human beings do not have or need any such rule. Our needs, desires, and emotions provide us directly with a sense of the appropriateness of our behavior. If these needs, desires, and emotions in turn depend on the abilities and vulnerabilities of a biological body socialized into a culture, even reinforcement-learning devices still have a very long way to go.

All work in AI, then, seems to face a deep dilemma. If one tries to build a GOFAI system, one finds that one has to represent in a belief system all that a human being understands simply by being a skilled human being. In my preface to the second edition of this book, the extreme unlikelihood of successfully programming the computer to show common sense by making explicit enough of what human beings understand simply by being embodied and skilled led me to skepticism concerning the GOFAI research program. Happily, recent research in machine learning does not require that one represent everything that human beings understand simply by being human. But then, as we have just seen, one encounters the other horn of the dilemma. One needs a learning device that shares enough human concerns and human structure to learn to generalize the way human beings do. And as improbable as it was that one could build a device that could capture our humanity in a physical symbol system, it seems at least as unlikely that one could build a device sufficiently like us to act and learn in our world.

Notes

1. Marvin Minsky, *Computation: Finite and Infinite Machines* (Englewood Cliffs, N.J.: Prentice-Hall, 1967), p. 2.

2. Gina Kolata, "How Can Computers Get Common Sense?," *Science*, Vol. 217, No. 24 (September 1982), p. 1237.

In their article "Could a Machine Think?" in *Scientific American* (January 1990), Paul and Patricia Churchland give an accurate overview of the role of this book in predicting and reinforcing this change of mood:

> In 1972 Hubert L. Dreyfus published a book that was highly critical of the parade-case simulations of cognitive activity. He argued for their inadequacy as simulations of genuine cognition, and he pointed to a pattern of failure in these attempts. What they were missing, he suggested, was the vast store of inarticulate background knowledge every person possesses and the commonsense capacity for drawing on relevant aspects of that knowledge as changing circumstance demands. Dreyfus did not deny the possibility that an artificial physical system of some kind might think, but he was highly critical of the idea that this could be achieved solely by symbol manipulation. . . .
>
> Dreyfus's complaints were broadly perceived within the AI community, and within the discipline of philosophy as well, as shortsighted and unsympathetic, as harping on the inevitable simplifications of a research effort still in its youth. These deficits might be real, but surely they were temporary. Bigger machines and better programs should repair them in due course. Time, it was felt, was on AI's side. . . .
>
> [But] time was on Dreyfus's side as well . . . . realistic performance required that the computer program have access to an extremely large knowledge base. Constructing the relevant knowledge base was problem enough, and it was compounded by the problem of how to access just the contextually relevant parts of that knowledge base in real time. As the knowledge base got bigger and better, the access problem got worse. Exhaustive search took too much time, and heuristics for relevance did poorly. Worries of the sort Dreyfus had raised finally began to take hold here and there even among AI researchers. [pp. 33–34]

3. For a more detailed account of the stages of skill acquisition, see Hubert Dreyfus and Stuart Dreyfus, *Mind Over Machine: The Power of Human*

Intuitive Expertise in the Era of the Computer (New York: Free Press, 1986). The occasion for writing this book was the enthusiasm for expert systems that had replaced the original hope for all-purpose intelligent machines. The idea was that programs that could exhibit intelligence in micro-worlds cut off from common sense, such as SHRDLU, could be adapted to deal with real-world micro-worlds such as blood disease diagnosis or spectrograph analysis. In this first wave of defections from the degenerating GOFAI program, theoretical AI researchers, now renamed knowledge engineers, predicted a race with Japan to develop intelligent machines that would contain all human expertise. *Newsweek* announced in a cover story that "Japan and the United States are rushing to produce a new generation of machines that can very nearly think." Japan's Fifth Generation Project assembled researchers from all of that country's computer companies to produce intelligent machines in ten years or, at least, as more cautious knowledge engineers said, to produce important spin-off results. On the basis of our analysis of expertise, my brother and I predicted defeat.

The ten years are up, and as the *Washington Post* has reported (June 2, 1992, p. C1), the Japanese government has closed the books on the project, conceding that it has had minimal impact on the global computer market. American designers of expert systems have also faced many disappointments. A typical case is the long-range project at Rockefeller University and Cornell University Medical College to develop computer programs that might help hematologists make diagnostic decisions. The researchers have already conceded defeat:

> Our efforts over many years in trying to develop a suitable diagnostic program for hematologic diseases confirms the great difficulty and even the impossibility of incorporating the complexity of human thought into a system that can be handled by a computer. . . .
>
> We do not see much promise in the development of computer programs to simulate the decision making of a physician. Well-trained and experienced physicians develop the uniquely human capability of recognizing common patterns from observation of exceedingly large numbers of stimuli. While people perform this intellectual synthesis quite readily, they are unable to spell out in detail how they do it; hence they cannot transfer the ability to a computer. Neither can they develop this ability, except in a very elementary way, by learning from computer-based programs intended to teach decision making.

4. One can, of course, recall the rules one once used and act on them again, but then one's behavior will be as halting and clumsy as it was when one followed the rules as a beginner.

5. Edward Feigenbaum and Pamela McCorduck, *The Fifth Generation: Artificial Intelligence and Japan's Computer Challenge to the World* (Reading, Mass.: Addison-Wesley, 1983).

6. Allen Newell and Herbert Simon, "Computer Science as Empirical Inquiry: Symbols and Search," in *Mind Design*, ed. John Haugeland (Cambridge, Mass.: MIT Press, 1985), p. 50.

7. D. O. Hebb, *The Organization of Behavior* (New York: Wiley, 1949).

8. The collapse of the GOFAI research program was accelerated by a loss of faith on the part of the U.S. Department of Defense, which had been its longtime supporter. The Defense Department, in what was perhaps a test of GOFAI research, announced in the early 1980s that it would support research aimed at three goals that AI researchers claimed could be reached within ten years, including a project to create a truck that could plot its own course and steer by itself. After five years and $500 million, none of the goals appeared close, and one by one the projects were closed down. Almost all Defense Department support for AI-related research now goes to neural-network projects.

9. Neural nets are being used in many ways other than what are called MLPs, but MLPs are easily the most frequently used procedure for decision-making. For our purposes we take "neural net" to be synonymous with MLP.

10. Paul Smolensky, "Connectionist AI, Symbolic AI, and the Brain," *Artificial Intelligence Review*, Vol. 1 (1987), p. 95.

11. Douglas Lenat, "Computer Software for Intelligent Systems," *Scientific American* (September 1984), p. 204.

12. Douglas Lenat and Edward Feigenbaum, "On the Thresholds of Knowledge," *Artificial Intelligence*, Vol. 47, Nos. 1–3 (January 1991), p. 199.

13. *Ibid.*, p. 216.

14. *Ibid.*, p. 188. Lenat has high hopes: "Our distant descendants may look back on the synergistic man-machine systems that emerge from AI as the natural dividing line between 'real human beings' and 'animals.' We stand, at the end of the 1980s, at the interstice between the first era of intelligent systems and the second era. . . . In that 'second era' of knowledge systems, the 'system' will be reconceptualized as a kind of colleagular relationship between intelligent computer agents and intelligent people" (pp. 224–225).

15. For more on these techniques, see p. 35 of this book.

16. Douglas Lenat and R. V. Guha, *Building Large Knowledge-Based Systems: Representation and Inference in the Cyc Project* (Reading, Mass.: Addison Wesley, 1990), p. 15.

17. Douglas B. Lenat et al., "CYC: Toward Programs with Common Sense," *Communications of the ACM*, Vol. 33, No. 8 (August 1990), p. 33.

18. Lenat and Guha, *Building Large Knowledge-Based Systems*, p. 23.

19. Lenat and Feigenbaum, "On the Thresholds of Knowledge," p. 200.

20. *Ibid.*

21. *Ibid.*, p. 218.

22. Oliver Sacks, *The Man Who Mistook His Wife for a Hat and Other Clinical Tales* (New York: Summit Books, 1986), p. 59.

23. Mark Johnson, *The Body in the Mind: The Bodily Basis of Meaning, Imagination, and Reason* (Chicago: The University of Chicago Press, 1987), p. 168.

24. Lenat and Feigenbaum, "On the Thresholds of Knowledge," p. 187.

25. *Ibid.*, p. 221.

26. Pierre Bourdieu, *Outline of a Theory of Practice* (Cambridge, England: Cambridge University Press, 1977), pp. 5–6.

27. Pierre Bourdieu, *The Logic of Practice* (Stanford: Stanford University Press, 1980), p. 54.

28. Lenat, "Computer Software for Intelligent Systems," p. 209.

29. Douglas Hofstadter, "Metafont, Metamathematics, and Metaphysics," *Visible Language*, Vol. 16 (April 1982). Hofstadter is interested in the various styles of type fonts. This is a case of style of a type of object, like clothing or speech. The pervasive style that interests Bourdieu and Merleau-Ponty, however, opens a disclosive space that allows anything to be encountered *as* something.

30. Lenat and Feigenbaum, "On the Thresholds of Knowledge," p. 201.

31. *Ibid.* (my italics).

32. Lenat and Guha, *Building Large Knowledge-Based Systems*, pp. 11–12.

33. Lenat and Feigenbaum, "On the Thresholds of Knowledge," p. 198.

34. John R. Searle, *Intentionality: An Essay in the Philosophy of Mind* (Cambridge, England: Cambridge University Press, 1983), pp. 95–96.

35. *Ibid.*, p. 149.

36. See, for example, Jerry R. Hobbes et al., *Commonsense Summer: Final Report*, CSLI Report No. 85-35, Center for the Study of Language and Information, Stanford University.

37. Lenat and Feigenbaum, "On the Thresholds of Knowledge," p. 219.

38. Of course, grandmasters do not make that move immediately. They first check out its effects on future events and also, time permitting, examine less plausible alternatives.

39. See Hubert L. Dreyfus, *Being-In-The-World: A Commentary on Heidegger's* Being and Time, *Division I* (Cambridge, Mass.: MIT Press, 1991).

40. Terry Winograd, "Heidegger and the Design of Computer Systems,"

speech delivered at the Applied Heidegger Conference, Berkeley, California, December 1989.

41. David Chapman, *Vision, Instruction, and Action* (Cambridge, Mass.: MIT Press, 1991), p. 20.

42. *Ibid.*, pp. 20–21.

43. *Ibid.*, p. 30.

44. *Ibid.*, p. 22.

45. When given additional text beyond the training samples, NETtalk's performance degrades noticeably, but it can still be understood by native speakers. Critics have pointed out, however, that this is not so much evidence that the network can generalize what it has learned to new texts as a demonstration of the ability of native speakers to make sense out of a stream of error-corrupted phonemes.

46. Furthermore, no correlations were found between *patterns* of activity over the hidden nodes and the input features. A consistent structure *was* found involving *relations* among patterns of activities among hidden nodes regardless of initial connection strengths. The structure, however, was not determined by the *input* to the net, as would be necessary if this structure were to be taken as a very abstract version of a theory of the domain. Rather, the structure involved not only the input letter to be pronounced but the correct pronunciation as well.

In our article in *Daedalus* (Winter 1988), Stuart and I asserted that the nodes of a particular net that had learned a particular skill could not be interpreted as having picked out stable, reidentifiable features of the skill domain. Our argument was that as soon as the net gave a wrong response and had to be corrected, the weights on its connections and the interpretation of its nodes would change. Thus the features the net picked out would be a function of what it had learned, not of the domain structure alone. But this argument begs the question. If the net *had* abstracted a theory of the domain, it would have converged on the correct features and so would not have had to be further corrected. Paul Churchland's analysis of NETtalk seems vulnerable to the same objection. See his chapter in Daniel Osherson and Edward E. Smith, eds., *Thinking*, vol. 3 of *An Invitation to Cognitive Science* (Cambridge, Mass.: MIT Press, 1990), p. 217.

47. I found this quoted in Michel Foucault, *The Order of Things* (New York: Vintage Books, 1973), p. xv.

48. For a statistical path to this conclusion, see S. Geman, E. Bienenstock, and R. Doursat, "Neural Networks and the Bias/Variance Dilemma," *Neural Computation*, Vol. 4, No. 1 (1992), pp. 1–58, esp. pp. 46–48. For a discussion of a psychological perspective on this realization, see R. N. Shepard, "Internal

Representation of Universal Regularities: A Challenge for Connectionism," in *Neural Connections, Mental Computation*, ed. Lynn Nadel et al. (Cambridge, Mass.: MIT Press, 1989), pp. 104–134.

49. Useful references on this subject are: *Machine Learning*, Vol. 8, Nos. 3/4 (May 1992), and A. G. Barto, "Reinforcement Learning and Adaptive Critic Methods," in *Handbook of Intelligent Control: Neural, Fuzzy and Adaptive Approaches*, ed. D. A. White and D. Sofge (New York: Van Nostrand Reinhold, 1992). In what follows we shall describe what might be called the minimal essence of such reinforcement learning. More complicated schemes using more derived information and optimization over alternatives at each decision point perform better on toy problems than what we shall describe but require much greater memory capacity and, more importantly, deviate from what seems to us to be phenomenologically plausible.

50. The technique can be seen as an asynchronous, successive-approximation version of the dynamic programming optimization procedure; what is being gradually learned, in the terminology of dynamic programming, is the optimal policy and the optimal value functions. In many real-world situations, there are no intermediate payoffs; the reinforcement comes only at the end of a process. In a game of chess, for example, most decisions (move choices) in a situation (board position) produce no reinforcement but only a transition to a new situation. Only a move terminating the game produces a positive, zero, or negative reinforcement. The device would nevertheless attempt to learn from experience a best value for each position (based on whether the position leads to a win, a draw, or a loss with perfect play) and a move that attains that value.

51. Gerald Tesauro, "Practical Issues in Temporal Difference Learning," *Machine Learning*, Vol. 8, Nos. 3/4 (May 1992), pp. 257–277.

52. When eight handcrafted features computed from the board position were added to its board description input features, the device learned to play at very close to grandmaster level and well above the level of any other program known to the net's creator.

53. To implement an automatic generalization procedure, one chooses some parameterized formulas and adjusts their parameters to give what are currently believed to be correct actions and values for situations in which such actions and values have been learned, and then one uses these formulas to produce actions and values in all cases. One often-used formula takes the form of a neural network in which the parameters are the connection strengths.

The backgammon program uses a network to produce its positional values. It avoids directly choosing actions by examining all possible actions in a situation and choosing the one that looks best on the basis of the situational values assigned by the algorithm. This goes beyond what we have called the minimal essence of reinforcement learning and does not fit the phenomenology of human intelligent action. As in the case of supervised-learning networks, there is nothing in the algorithm that guarantees that it will generalize correctly when applied to a new situation. For this very reason, its success surprised its designer. It may well be that, as in the case of NETtalk, this seemingly successful generalization merely shows that in this domain accuracy in generalization is not essential.

54. David Chapman and Leslie Pack Kaelbling, "Input Generalization in Delayed Reinforcement Learning: An Algorithm and Performance Comparisons," *Proceedings of the 1991 International Joint Conference on Artificial Intelligence* (Cambridge, Mass.: AAAI Press/MIT Press, 1991), pp. 726–731.